



SpiderVault 20.0

Model: SpiderVault 20.0 (3P/3P) · 20 kVA Three-Phase UPS / Solar Hybrid

INPUT	
Voltage	3 × 415 VAC
Selectable Voltage Range	380 – 440 VAC
Frequency Range	50 Hz / 60 Hz Auto Sensing

OUTPUT	
AC Voltage Regulation (Battery Mode)	415 VAC ± 5%
Surge Power	36,000 VA
Efficiency (Peak)	>97%
Switching Modes	On Grid Off Grid Hybrid (Optional)
Transfer Time	10 ms
Waveform	Pure Sine Wave

SOLAR CHARGER & AC CHARGER			
Solar Charger Type	MPPT	Number of MPPTs	3
Max. PV Array Open Circuit Voltage	450 VDC	Maximum PV Array Power	3 × 8,500 W
Maximum PV Input Current	3 × 27 A	MPP Range @ Operating Voltage	90 – 450 VDC
Maximum Solar Charge Current	360 A	Maximum AC Charge Current	360 A

BATTERY	
Cell Technology	Lithium Ion
Nominal Voltage	48 V
Battery Capacity	20 kWh
Max. Continuous Charging Current	40 A
Max. Continuous Discharging Current	90 A × 5
Full Charging Voltage	54.3 V
Full Discharging Voltage	40 V

PROTECTION	
Protection System	5th Gen BMS DC Breaker
Anti-islanding Protection	Integrated
AC Over Current Protection	Integrated
Output Short Circuit	Integrated
Overload Protection	Integrated
Battery Voltage Too Low	Integrated
Temperature Too High	Integrated
Input Voltage Out of Range / Ripple	Integrated
Fan Block	Integrated
IP Protection	IP65

GENERAL DATA	
Dimensions W × D × H	540 × 490 × 1325 mm
Net Weight	210 kg
Installation Style	Floor Standing
Cooling Concept	Advanced Nano PCM + Forced Convection
Operating Temperature Range	–10°C to 60°C
Noise Level	< 50 dB
Topology	High Frequency Isolation
LCD Display	Operating Mode / Load / Input / Output
PV Connection	MC4 Connector
AC Connection	Terminal Connectors
No. of AC Outputs	1
Communication Interface	BMS Dry-contact CAN Bluetooth Wi-Fi AI Cloud
Warranty	5 Years (Extended: 12 Years)

✓

CERTIFICATIONS

- EN-IEC 60335-1
- EN-IEC 60335-2-29
- IEC 62109-1

20 kW

Rated Power

>97%

Peak Efficiency

20 kWh

Battery Capacity

10 ms

Transfer Time

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